

Topic 1-practice

Daniel Trad

Pressure and Displacement Formulations

Consider acoustic wave propagation in a homogeneous fluid.

The wavefield may be described either by the scalar pressure field

$$\frac{\partial^2 p}{\partial t^2} = v^2 \nabla^2 p,$$

or by the particle displacement vector $\mathbf{u}(\mathbf{x}, t) = \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix}$.

Given information

Assume that pressure is related to displacement through

$$p = -K \nabla \cdot \mathbf{u},$$

where K is the bulk modulus, and $\nabla \cdot \mathbf{u} = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}$. Also assume that each displacement component satisfies

$$\frac{\partial^2 u_i}{\partial t^2} = v^2 \nabla^2 u_i, \quad i = x, y, z.$$

Discussion Questions

Discuss the following:

1. Why is p a **scalar** field, while $\mathbf{u}$ requires **three components** in 3D?
2. Take the divergence of the three displacement equations. Show that $\nabla \cdot \mathbf{u}$ satisfies a scalar wave equation, and therefore so does p .
3. What information about the displacement field is retained by $\nabla \cdot \mathbf{u}$, and what information is lost?
4. Why is pressure often a more economical variable for describing acoustic waves in a fluid?
5. Does knowing $p(\mathbf{x}, t)$ uniquely determine all three components u_x, u_y, u_z ? Explain.

Conclusion to discuss

In what sense can one scalar pressure equation replace three displacement equations, and in what sense are the two formulations not completely identical?

Instructor Solution / Sketch

1. **Scalar vs. vector:** Pressure p has one value at each point, so it is scalar. Displacement $\mathbf{u} = (u_x, u_y, u_z)$ has three components, so it is vector-valued.
2. **Take the divergence of the displacement equations:**

$$\frac{\partial^2 u_i}{\partial t^2} = v^2 \nabla^2 u_i \quad (i = x, y, z).$$

Apply $\nabla \cdot$:

$$\frac{\partial^2}{\partial t^2} (\nabla \cdot \mathbf{u}) = v^2 \nabla^2 (\nabla \cdot \mathbf{u}).$$

Since

$$p = -K \nabla \cdot \mathbf{u},$$

and K is constant, it follows that

$$\frac{\partial^2 p}{\partial t^2} = v^2 \nabla^2 p.$$

Instructor Solution / Sketch

3. **What is retained / lost?** The divergence keeps the **compressional (volumetric)** part of motion. It loses the full directional information of $\mathbf{u}$, especially any part not seen in $\nabla \cdot \mathbf{u}$.
4. **Why pressure is economical:** For acoustic waves in fluids, the main physics is compressional, so one scalar p is often enough instead of three unknowns u_x, u_y, u_z .
5. **Important subtlety:** Knowing p does *not* generally determine the full displacement vector uniquely. Pressure determines $\nabla \cdot \mathbf{u}$, not all components of $\mathbf{u}$.

Reflection at the Free Surface and Multiples

Consider an upgoing primary wave reaching the Earth's surface. The pressure reflection coefficient for normal incidence is

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1},$$

where $Z = \rho v$ is the acoustic impedance. At the free surface, $Z_{\text{air}} = \rho_{\text{air}} v_{\text{air}} \ll Z_{\text{earth}}$, and therefore $R_{\text{surface}} = \frac{Z_{\text{air}} - Z_{\text{earth}}}{Z_{\text{air}} + Z_{\text{earth}}} \approx -1$.

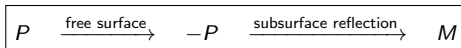
Physical interpretation

If the upgoing primary arriving at the surface has amplitude P ,

$$P_{\text{reflected}} = R_{\text{surface}} P \approx -P.$$

Thus, the wave reflected back into the Earth has approximately **the same amplitude as the arriving primary**, but with **opposite polarity**.

This downgoing wave can reflect again from subsurface interfaces and return to the receivers as a **free-surface multiple**:



Therefore, the free surface does not significantly attenuate the wave: it sends essentially the full primary amplitude back into the Earth, allowing strong multiples to be generated.